# Factorization memory in a conic ideal: the $A_3$, $B_3$, and $H_3$ configurations

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**Abstract**

Restriction to a conic forgets how its marked points were paired into secants. We study the information retained after differences of matching products are divided by the conic equation. We prove that this quotient is intrinsic up to translation and equivariant under the endpoint stabilizer. For the rank-three Coxeter configurations $A_3$, $B_3$, and $H_3$, its difference space has ranks 3, 6, and 10. Uniformly, the image contains the top harmonic summand and, in even degree, the deepest radial line. In the $H_3$ case it omits exactly the middle Fischer layer $Q\mathcal{H}_2$. In the balanced cases, second moments recover two complementary sheets, a signed cubic orients them, and, relative to a selected $A_4$ refinement, six incidence profiles recover the matching-table row. A modular depth quotient and arithmetic gluing describe how this factorization memory changes across characteristics.

**Keywords.** conic ideal; secant matching; Coxeter configuration; factorization; cubic invariant; modular representation.

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## 1 Introduction

The companion rigidity paper isolates the Clebsch hexagon through its uncovered syndrome locus and reconstructs its projective geometry from decoding data [5]. Here we begin with marked secant geometry and ask a complementary reconstruction question: what does common restriction to a conic forget, and what does the conic-ideal quotient remember? We answer this question across the rank-three $A_3$, $B_3$, and $H_3$ configurations.

**Theorem 1.1** (Factorization recovery). *Let the $A_3$, $B_3$, and $H_3$ matching configurations be the finite matching orbits defined in Section 4. Then:*

(i) *canonically scaled matching products have equal restriction to the conic, and their conic-ideal quotients have difference ranks* 3, 6, *and* 10*;*

(ii) *in the $B_3$ and $H_3$ cases, the two factorization sheets are, up to interchange, the unique complementary halves with equal first and second tensor moments;*

(iii) *the signed moments vanish through degree two, while a nonzero degree-three tensor transforms by the character that exchanges the sheets;*

(iv) *in the $H_3$ case, six explicit incidence profiles separate the six $A_4$–$A_5$ double-coset labels, and the singleton profiles recover the corresponding decorated matching-table row;*

(v) *over* $\mathbb{F}_{11}$*, the linearized depth map is the* $A_4$*-fixed part of the projective cover of the trivial* $\mathrm{PSL}_2(11)$*-module, modulo its socle;*

(vi) *the* $A_3$ *marker is arithmetically fused over* $\mathbb{F}_5$*, whereas the* $B_3$ *and* $H_3$ *markers split into two rational sheets over* $\mathbb{F}_7$ *and* $\mathbb{F}_{11}$*; in the* $H_3$ *case the two* $A_5$ *stabilizers glue along* $A_4$ *and generate* $\mathrm{PSL}_2(11)$.

The theorem follows one mechanism from a four-endpoint switch to the recovery of finite matching data. Division by the conic equation first turns different factorizations of one restricted form into an affine configuration. Quadratic products then recover an unordered bipartition; the first signed moment that survives is cubic. Finally, a double-coset incidence map compresses the configuration to six profiles without identifying any two labels. The rigidity theorem motivates this question but is not a hypothesis: all marked-conic objects and quotient constructions are defined independently here.

This line of research arose from the following normal-play placement game. Two players alternately adjoin a point of a finite projective plane while keeping the selected set an arc; the first player with no legal point loses. After frame reduction in an odd projective plane, a residual three-point position and the two burned directions determine a five-arc and hence a conic; adjoining an on-conic response exposes the exceptional six-arc geometry studied here. The resulting connection now runs in both directions. The small-arc equality theorem isolates the projective frame over $\mathbb{F}_5$ and the Clebsch hexagon over $\mathbb{F}_{11}$ as the only configurations of sizes four through seven whose uncovered locus is a full conic. The latter yields an icosahedral continuation complex, a certified game position, and a symmetry-compressed finite model for constructing second-player responses. Under the standard arc–MDS–AME dictionaries [8], the same $H_3$ six-arc gives an $[6,3,4]_{11}$ MDS code and a minimum-support stabilizer absolutely maximally entangled state on six 11-level systems, so its projective rigidity controls a concrete quantum-code input. More generally, prescribed-hole arc theory places its equality and reconstruction properties in a forward/inverse framework, while the syndrome–uncovered-point mechanism extends to projective Reed–Solomon deep holes beyond redundancy four. These are applications and continuations of the geometry, not hypotheses of the results proved here [6, 7].

## 2 Conic matching products

Write the standard conic as

$$\mathcal{C} = V(Q), \qquad Q = XZ - Y^2,$$

and parametrize it by $\nu(s:t) = (s^2 : st : t^2)$. For points $a = (a_0 : a_1)$ and $b = (b_0 : b_1)$ of $\mathbb{P}^1$, put $[a,b] = a_0b_1 - a_1b_0$. We scale the secant $L_{ab}$ through $\nu(a)$ and $\nu(b)$ by the condition

$$L_{ab}(\nu(s:t)) = [a,(s:t)]\,[b,(s:t)].$$

Let $\Omega$ be a set of $2m$ distinct marked points of $\mathbb{P}^1$, with fixed vector representatives. For a perfect matching $M$ of $\Omega$, define its matching product

$$P_M = \prod_{\{a,b\}\in M} L_{ab}.$$

The vector representatives are part of the marked-conic data. Their role is to make the products comparable before passing to projective equivalence.

**Proposition 2.1** (The matching-secant quotient)**.** *For every $m \geq 2$ and every two perfect matchings $M, N$ of $\Omega$,*

$$P_M|_{\mathcal{C}} = P_N|_{\mathcal{C}}, \qquad P_M - P_N \in (Q).$$

*Consequently, after choosing a reference matching $M_0$,*

$$\Phi_{M_0}(M) = \frac{P_M - P_{M_0}}{Q} \in H^0(\mathbb{P}^2, \mathcal{O}(m-2)) \tag{2.1}$$

*is an affine configuration. Changing the reference matching translates this configuration. If $g \in \mathrm{GL}_2$ preserves $\Omega$, $\rho(g)$ is its action on binary quadratics, and representatives are transported by $g$, then*

$$\Phi_{gM_0}(gM) \circ \rho(g) = \det(g)^{2m-2}\Phi_{M_0}(M). \tag{2.2}$$

*Thus the difference space, its rank, and its affine moment conditions do not depend on $M_0$ and are equivariant under the endpoint stabilizer.*

*Proof.* For $x \in \mathbb{P}^1$,

$$P_M(\nu(x)) = \prod_{a \in \Omega} [a, x],$$

which does not depend on the pairing. The kernel of the Veronese pullback

$$k[X, Y, Z] \longrightarrow k[s, t], \qquad (X, Y, Z) \longmapsto (s^2, st, t^2)$$

is the principal ideal $(Q)$, so $Q$ divides every difference. Changing the reference follows by subtraction. Finally, $L_{ga,gb} \circ \rho(g) = \det(g)^2 L_{ab}$ and $Q \circ \rho(g) = \det(g)^2 Q$, which gives (2.2). □

# 3 Switches and divisibility

For four distinct endpoints, the Plücker relation gives the local switch identity

$$L_{ab}L_{cd} - L_{ac}L_{bd} = [a, d][b, c]\, Q. \tag{3.1}$$

Multiplying (3.1) by the secant factors common to two matchings proves divisibility for a single switch. Any two perfect matchings are joined by a sequence of such switches: if $M$ and $N$ differ at $a$, write $\{a, b\} \in M$ and $\{a, c\} \in N$; switching the two $M$-edges through $b$ and $c$ creates $\{a, c\}$ and reduces the symmetric difference. This also gives an elementary proof of the divisibility clause in Proposition 2.1.

The switch calculation and the global pullback proof play different roles. The pullback proves the quotient for arbitrary matchings at once. The switch identity records how local changes of factorization move in the quotient space. A selected Coxeter orbit need not contain every perfect matching or every intermediate switch.

# 4 The rank-three configurations

We now select three finite matching orbits. This selection is discrete input: Proposition 2.1 applies to every matching, but it does not distinguish the Coxeter orbits.

For $q \in \{5, 7, 11\}$, index

$$\mathbb{P}^1(\mathbb{F}_q) = \{0, 1, \ldots, q-1, \infty\},$$

where $a$ denotes $(1:a)$ and $\infty$ denotes $(0:1)$. Let $G_q = \mathrm{PGL}_2(q)$ act in the usual way. The three base matchings are

$$\begin{array}{ccl} T & q & M_T \\ \hline A_3 & 5 & \{0,\infty\},\{1,4\},\{2,3\}, \\ B_3 & 7 & \{0,2\},\{1,4\},\{3,\infty\},\{5,6\}, \\ H_3 & 11 & \{0,1\},\{2,5\},\{3,7\},\{4,9\},\{6,8\},\{10,\infty\}. \end{array} \tag{4.1}$$

Define the *rank-three matching configuration*

$$\Omega_T = G_q \cdot M_T.$$

The stabilizers of the three displayed matchings are respectively $S_4, S_4, A_5$, and hence

$$|\Omega_{A_3}| = 5, \qquad |\Omega_{B_3}| = 14, \qquad |\Omega_{H_3}| = 22. \tag{4.2}$$

These marker spaces and their stabilizers are classical. In particular, Edge and Dye identify the exceptional finite conic configurations and substantial parts of their incidence geometry [2, 3]. We use the Coxeter names for the corresponding tetrahedral, octahedral, and icosahedral data; neither the orbit counts in (4.2) nor the ambiguity of an unmarked conic is asserted here as new.

Fix $M_T$ as reference and apply (2.1) to every member of $\Omega_T$. Write $R_d = \mathbb{F}_q[X,Y,Z]_d$ and

$$\Delta_Q = 4\partial_X\partial_Z - \partial_Y^2, \qquad \mathcal{H}_d = \ker(\Delta_Q : R_d \longrightarrow R_{d-2}). \tag{4.3}$$

The operator $\Delta_Q$ is the Laplacian attached to the dual quadratic form. In the degrees used below, $\mathcal{H}_d$ is the top harmonic summand, while powers of $Q$ give radial summands.

**Theorem 4.1** (The $3,6,10$ quotient ranks)**.** *For the matching configurations in (4.1), let*

$$W_T = \mathrm{span}_{\mathbb{F}_q}\{\Phi_{M_T}(M) : M \in \Omega_T\}.$$

*Then*

$$\begin{array}{cccc} T & \deg\Phi & W_T & \dim W_T \\ \hline A_3 & 1 & R_1 = \mathcal{H}_1 & 3, \\ B_3 & 2 & R_2 = \mathcal{H}_2 \oplus \mathbb{F}_7 Q & 6, \\ H_3 & 4 & \mathcal{H}_4 \oplus \mathbb{F}_{11} Q^2 & 10. \end{array} \tag{4.4}$$

*If $h_T = 4, 6, 10$ is the Coxeter number and $d = h_T/2 - 1$, the three rows have the uniform form*

$$W_T = \begin{cases} \mathcal{H}_d, & d \text{ odd}, \\ \mathcal{H}_d \oplus \mathbb{F}_q Q^{d/2}, & d \text{ even}, \end{cases} \qquad \dim W_T = h_T - 1 + \mathbf{1}_{\{d \text{ even}\}}.$$

*In particular, the first two configurations fill the complete quotient form space. The $H_3$ configuration spans a canonical ten-dimensional subspace of the fifteen-dimensional space $R_4$.*

*Proof mode.* The quotient construction and the harmonic-space dimensions are proved symbolically. The assertions that the three displayed finite orbits span the stated subspaces are certificate-checked exact calculations, with a separate implementation that reconstructs the groups, matchings, quotient forms, Laplacian, and row reductions from the data in (4.1).

*Proof.* A matching in (4.1) has $m = 3, 4, 6$ edges, respectively, so Proposition 2.1 places its quotient in $R_{m-2}$, of degrees $1, 2, 4$. Representing the quotients in the standard monomial bases gives matrices with respectively $5, 14, 22$ rows. Exact row reduction over $\mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}$ gives ranks $3, 6, 10$.

It remains to identify the ten-space in the last row rather than record only its dimension. Direct differentiation gives

$$\dim \mathcal{H}_1 = 3, \qquad \dim \mathcal{H}_2 = 5, \qquad \dim \mathcal{H}_4 = 9.$$

The radial vectors $Q \in R_2$ and $Q^2 \in R_4$ are not harmonic in characteristics 7 and 11, respectively. The exact quotient matrices lie in

$$\mathcal{H}_1, \qquad \mathcal{H}_2 + \mathbb{F}_7 Q, \qquad \mathcal{H}_4 + \mathbb{F}_{11} Q^2.$$

Their computed ranks equal the dimensions $3, 6, 10$ of these spaces, so all three containments are equalities. Since $\dim R_4 = 15$, the last equality is proper. □

**Corollary 4.2** (The exact $H_3$ loss)**.** *Over $\mathbb{F}_{11}$, the quartic form space has the Fischer decomposition*

$$R_4 = \mathcal{H}_4 \oplus Q\mathcal{H}_2 \oplus \mathbb{F}_{11} Q^2. \tag{4.5}$$

*Consequently,*

$$R_4 = W_{H_3} \oplus Q\mathcal{H}_2, \qquad R_4/W_{H_3} \simeq Q\mathcal{H}_2. \tag{4.6}$$

*Equivalently,*

$$W_{H_3} = \{f \in R_4 : \Delta_Q f \in \mathbb{F}_{11} Q\}. \tag{4.7}$$

*Thus the $H_3$ quotient configuration retains the top harmonic nine-space and the deepest radial line, and omits precisely the five-dimensional middle radial layer.*

*Proof.* For a homogeneous form $f$ of degree $r$, direct differentiation gives

$$\Delta_Q(Qf) = (4r + 6)f + Q\Delta_Q f. \tag{4.8}$$

Hence $\Delta_Q(Qh) = 3h$ for $h \in \mathcal{H}_2$, while $\Delta_Q(Q^2) = 9Q$ over $\mathbb{F}_{11}$. Applying $\Delta_Q^2$ and then $\Delta_Q$ shows that the three summands in (4.5) meet trivially. Their dimensions are $9, 5, 1$, so they exhaust the fifteen-dimensional space $R_4$. Theorem 4.1 then gives (4.6). If $f = h_4 + Qh_2 + cQ^2$ is its decomposition, then

$$\Delta_Q f = 3h_2 + 9cQ.$$

The quadratic decomposition $R_2 = \mathcal{H}_2 \oplus \mathbb{F}_{11} Q$ therefore gives (4.7). □

The rank calculation has two distinct boundaries. The common restriction and quotient are general consequences of the conic ideal, and (4.3) describes the intrinsic target subspaces. The fact that the three particular orbits in (4.1) attain those subspaces is finite. The primary certificate enumerates each full orbit and performs exact prime-field row reduction; the independent replay starts only from (4.1) and reconstructs the calculation without importing the primary implementation. Thus Theorem 4.1 does not infer a general rank formula from the three examples.

Equation (4.7) isolates a smaller conceptual target than the full rank calculation: the $H_3$ containment is the vanishing of the harmonic quadratic part of $\Delta_Q \Phi(M)$. Once that covariant identity is known, only spanning the resulting ten-space remains finite.

Corollary 4.2 identifies the linear information lost by the $H_3$ orbit, but linear span does not recover the two factorization sheets inside the finite configuration. Their recovery begins one categorical level higher, with products of affine evaluation functions.

## 5 Balanced sheets and cubic orientation

For $T = B_3, H_3$, put $q = 7, 11$, respectively. The subgroup

$$G^+ = \mathrm{PSL}_2(q) \subset G = \mathrm{PGL}_2(q)$$

splits $\Omega_T$ into two orbits $\Omega_+ \sqcup \Omega_-$, each of size $q$; an element of the outer coset exchanges them. These are the two one-factorization sheets. The $A_3$ orbit does not split: its $S_4$ stabilizer cannot lie in $\mathrm{PSL}_2(5) \simeq A_5$, already by order.

Write $x_M = \Phi_{M_T}(M) \in W_T$. Let $L \subseteq \mathbb{F}_q^{\Omega_T}$ be the evaluation space of affine-linear functions on the finite configuration $\{x_M\}$. Since the points affinely span $W_T$,

$$\dim L = 1 + \dim W_T = q.$$

For subspaces of functions, write

$$L^{\circ 2} = \mathrm{span}\{fg : f, g \in L\},$$

where multiplication is pointwise. The common second-moment form in the proposition below is

$$B(\lambda, \mu) = \sum_{M \in \Omega_+} \lambda(x_M)\mu(x_M) = \sum_{M \in \Omega_-} \lambda(x_M)\mu(x_M)$$

on $W_T^\vee$. A *field-valued strength-two trade* is a weight $\tau \in k^\Omega$ satisfying, for all $f, g \in L$,

$$\sum_{M \in \Omega} \tau(M) f(M) g(M) = 0.$$

**Proposition 5.1** (Radical–Hadamard recovery)**.** *Let $k$ be a field and $\Omega = \Omega_+ \sqcup \Omega_-$, with both sheets of size $q = 0$ in $k$. Suppose an affine evaluation space $L \subseteq k^\Omega$ has the following properties:*

(i) *each sheet has zero first moment, and the two sheets have equal second moments;*

(ii) *$\dim L = q$, and restriction from $L$ onto the zero-sum hyperplane of either sheet has rank $q - 1$;*

(iii) *the second-moment form on linear functionals has rank $q - 2$ and a one-dimensional radical;*

(iv) *a nonzero radical functional is constant on each sheet, with different constants on the two sheets.*

*Then*

$$L^{\circ 2} = \left\{ (u_+, u_-) : \sum_{\Omega_+} u_+ = \sum_{\Omega_-} u_- \right\}, \tag{5.1}$$

*and*

$$(L^{\circ 2})^\perp = k(\mathbf{1}_{\Omega_+} - \mathbf{1}_{\Omega_-}). \tag{5.2}$$

*Consequently every field-valued strength-two trade is a scalar multiple of the sheet sign.*

*Proof.* Let $e_+$ and $e_-$ be the two sheet indicators. The radical functional evaluates as $a_+e_+ + a_-e_-$ with $a_+ \neq a_-$. Together with $1 = e_+ + e_- \in L$, it therefore yields $e_+, e_- \in L$.

Let $H_\pm$ be the zero-sum hyperplane on $\Omega_\pm$. The first-moment hypothesis and $q = 0$ put each restriction of $L$ inside $H_\pm$; the rank hypothesis makes both restrictions surjective. Multiplication by $e_+$ and $e_-$ now gives

$$(H_+ \oplus 0) + (0 \oplus H_-) \subseteq L^{\circ 2}. \tag{5.3}$$

This subspace has dimension $2q - 2$.

For $f, g \in L$, the difference between the two sheet sums of $fg$ expands into constant, first-moment, and second-moment terms. The hypotheses make all three differences zero, so $L^{\circ 2}$ lies in the hyperplane on the right of (5.1). Finally choose restrictions $a, b \in H_+$ with nonzero standard pairing and lift them to $f, g \in L$. Equality of the second moments gives the same nonzero pairing on $\Omega_-$. Thus $fg$ has equal nonzero sheet sums and does not lie in (5.3). We have found $2q - 1$ independent directions in the $2q - 1$-dimensional hyperplane, proving (5.1). Its annihilator under the standard coordinate pairing is the line in (5.2). □

The hypotheses of Proposition 5.1 are small finite linear-algebra statements for the two quotient configurations:

| $T$ | $q$ | sheet restriction ranks | second-moment rank | $\dim L^{\circ 2}$ |
|---|---|---|---|---|
| $B_3$ | 7 | $6, 6$ | 5 | 13, |
| $H_3$ | 11 | $10, 10$ | 9 | 21. |

(5.4)

In each row the radical is one-dimensional and takes two distinct sheet values. The exact certificate checks these hypotheses from the matching quotients; an independent implementation reconstructs the row reductions and the resulting trade line. No enumeration of equal-size subsets is used in the proof.

**Theorem 5.2** (Balanced sheets and cubic-first orientation)**.** *For $T = B_3, H_3$, the unordered pair $\{\Omega_+, \Omega_-\}$ is intrinsic to the affine quotient configuration:*

(i) *up to interchange, $\Omega_+$ and $\Omega_-$ are the only complementary halves with equal first and second tensor moments;*

(ii) *if $\epsilon$ is $+1$ on $\Omega_+$ and $-1$ on $\Omega_-$, then*

$$\mu_j = \sum_{M \in \Omega_T} \epsilon(M) x_M^{\odot j} \in \mathrm{Sym}^j(W_T)$$

*satisfies*

$$\mu_1 = 0, \qquad \mu_2 = 0, \qquad \mu_3 \neq 0; \tag{5.5}$$

(iii) *$\mu_3$ is independent of the reference matching, $G^+$ fixes it, and the outer coset negates it. Within $G$,*

$$\mathrm{Stab}_G(\mu_3) = G^+, \qquad \mathrm{Stab}_G(\mathbb{F}_q \mu_3) = G. \tag{5.6}$$

*Thus degree three is the first signed tensor moment, or linear functional of such a moment, that orients the recovered sheet pair. This minimality statement concerns signed tensor moments, not every possible nonlinear statistic of the configuration.*

*Proof mode.* Proposition 5.1 is a symbolic proof. The ranks, radical separation, lower-moment cancellation, and nonzero cubic for the two displayed configurations are certificate-checked over $\mathbb{F}_7$ and $\mathbb{F}_{11}$ and independently replayed. The transformation law and reference independence below are formal consequences of those finite inputs.

*Proof.* Equality of first and second tensor moments for a complementary pair is equivalent, by polarization, to its sign weight annihilating $L^{\circ 2}$. Since the characteristics are 7 and 11, polarization through degree two is valid. Proposition 5.1 says that the annihilator is exactly the sheet-sign line. A sign weight taking only the values $\{\pm 1\}$ is therefore $\epsilon$ or $-\epsilon$, proving (i).

The same moment equalities give $\mu_1 = \mu_2 = 0$. Exact tensor summation in the rank-six and rank-ten quotient coordinates gives $\mu_3 \neq 0$. If the reference matching changes, every point translates by one vector $c$. Expanding the signed third moment after $x_M \mapsto x_M + c$ leaves $\mu_3$ unchanged: the other terms contain $\mu_2, \mu_1$, or $\mu_0 = \sum_M \epsilon(M) = 0$.

Every element of $G^+$ preserves both sheets and hence fixes the signed sum. Every element of the outer coset exchanges the sheets and negates it. Since $\mu_3 \neq 0$, these two transformation laws give (5.6). Finally (5.5) proves the stated minimality inside the signed tensor-moment family. □

The cubic remembers an orientation that the linear quotient does not. Indeed, the signed linear relation

$$\sum_{M \in \Omega_+} x_M = \sum_{M \in \Omega_-} x_M$$

already shows that the sheet sign is killed by the linear quotient map. Quadratic products recover the unordered partition through (5.2), and the cubic supplies the sign exchanged when the two recovered levels are swapped.

**Corollary 5.3** (Secant-product tensor syzygies)**.** *For $T = B_3, H_3$, let $m_Q : R_{m-2} \to R_m$ denote multiplication by $Q$. Regarding the displayed powers as symmetric tensors, not polynomial powers, the canonically scaled secant products satisfy*

$$\begin{aligned} &\sum_M \epsilon(M) P_M = 0, \\ &\sum_M \epsilon(M) P_M^{\odot 2} = 0, \\ &\sum_M \epsilon(M) P_M^{\odot 3} = \mathrm{Sym}^3(m_Q)(\mu_3) \neq 0. \end{aligned} \tag{5.7}$$

*Proof.* Substitute $P_M = P_{M_T} + Q x_M$ and expand. Each sheet has $q = 0$ elements in $\mathbb{F}_q$, and the signed quotient moments vanish through degree two. All terms below cubic degree cancel, leaving $\mathrm{Sym}^3(m_Q)(\mu_3)$ in degree three. Since $\mathbb{F}_q[X, Y, Z]$ is a domain, multiplication by $Q$ is injective; over a field, the symmetric powers of an injective linear map are injective. □

# 6 Six profiles and matching-row reconstruction

We now specialize to $H_3$. Put

$$G = \mathrm{PGL}_2(11), \qquad G^+ = \mathrm{PSL}_2(11), \qquad X = \Omega_{H_3} \simeq G/H,$$

where $H \simeq A_5$ is the stabilizer of the base matching in (4.1). Thus $X = X_+ \sqcup X_-$, with $|X_+| = |X_-| = 11$. Here is the additional decoration used in this section. On $V = \mathbb{F}_{11}^3$, set

$$q_0(x, y, z) = x^2 + y^2 + z^2, \qquad \mathcal{C}_0 = \mathbb{P}\{q_0 = 0\}.$$

The projectivity

$$\psi = \begin{pmatrix} 10 & 2 & 1 \\ 1 & 2 & 10 \\ 3 & 0 & 3 \end{pmatrix} \quad \text{satisfies} \quad q_0(\psi(X,Y,Z)^t) = 3(XZ - Y^2), \tag{6.1}$$

so $\psi \circ \nu$ identifies the endpoint conic of Section 2 with $\mathcal{C}_0$. In these coordinates take the two six-point sets

$$\begin{aligned} A_+ &= \{(0:1:4),(0:1:7),(1:0:3),(1:0:8),(1:4:0),(1:7:0)\}, \\ A_- &= \{(0:1:3),(0:1:8),(1:0:4),(1:0:7),(1:3:0),(1:8:0)\}. \end{aligned} \tag{6.2}$$

Let $\mathcal{P} = G \cdot A_+$, where $G$ is viewed as the full projective stabilizer of $\mathcal{C}_0$. For $A \in \mathcal{P}$ and $a \in A$, let $C_{A,a}$ be the conic through $A \setminus \{a\}$. The six pairs

$$C_{A,a} \cap \mathcal{C}_0, \qquad a \in A, \tag{6.3}$$

form a perfect matching $M_A$ of the twelve points of $\mathcal{C}_0$. On this 22-point orbit, $A \mapsto M_A$ is a $G$-equivariant bijection onto $X$. This is the *matching-decorated parent correspondence*; its bijectivity and the assertion that (6.3) consists of six disjoint pairs are exact finite inputs.

Put $H_\pm = \mathrm{Stab}_G(A_\pm)$ and

$$K = H_+ \cap H_-, \qquad J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}. \tag{6.4}$$

Then $K \simeq A_4$, $J(A_+) = A_-$, and $J$ normalizes $K$ and exchanges $X_+$ with $X_-$. Thus the *ordered golden pair* means precisely the ordered pair $(A_+, A_-)$.

Choose linear representatives of $K$ preserving $q_0$, and let $\widetilde{K} = \mathbb{F}_{11}^\times K$. For the following eight vectors, define $R_i = \mathbb{P}(\widetilde{K} r_i)$:

| $i$ | $r_i$ | $i$ | $r_i$ |
|---|---|---|---|
| 1 | $(0,1,4)$ | 10 | $(0,1,3)$ |
| 3 | $(0,1,2)$ | 13 | $(0,1,5)$ |
| 6 | $(1,3,2)$ | 14 | $(1,1,5)$ |
| 9 | $(1,4,2)$ | 11 | $(1,2,4)$. |

(6.5)

These are cells in the scalar-$K$ orbit partition of $\mathbb{P}(V)$. The involution $J$ exchanges the four pairs

$$(R_1, R_{10}), \quad (R_3, R_{13}), \quad (R_6, R_{14}), \quad (R_9, R_{11}). \tag{6.6}$$

Only these four oriented pairs enter the argument. Reversing the golden pair reverses all four orientations.

This profile layer uses more decoration than the balanced-sheet theorem. The quotient configuration intrinsically recovers the unordered sheets, but it does not select an ordered golden pair or the resulting scalar-$K$ refinement. The reconstruction below is relative to that selected refinement.

For $M \in X$, let $S_M$ be the union of its six secants in $\mathbb{P}^2(\mathbb{F}_{11})$, and put

$$Z_M(R) = |S_M \cap R|.$$

Equivalently, $Z_M(R)$ counts the points of $R$ at which $P_M$ vanishes. Define the *depth profile*

$$\begin{aligned} D(M) = (&Z_M(R_1) - Z_M(R_{10}),\ Z_M(R_3) - Z_M(R_{13}), \\ &Z_M(R_6) - Z_M(R_{14}),\ Z_M(R_9) - Z_M(R_{11})) \in \mathbb{F}_{11}^4. \end{aligned} \tag{6.7}$$

This definition uses only zero sets of secant products, so it is unchanged if their equations are rescaled.

**Theorem 6.1** (Six-profile reconstruction)**.** *The $K$-orbits on each sheet $X_\pm$ have sizes $1, 4, 6$. With the orientation in (6.6), the corresponding profiles and fibre sizes are*

$$\begin{array}{ccc} \textit{sheet} & |D^{-1}(v)| & v \\ \hline + & 1 & v_1 = (-6, 0, 12, -12), \\ + & 4 & v_2 = (-3, 3, 0, 3), \\ + & 6 & v_3 = (3, -2, -2, 0), \\ - & 1 & -v_1, \\ - & 4 & -v_2, \\ - & 6 & -v_3. \end{array} \tag{6.8}$$

*The six vectors are pairwise distinct. Hence $D$ recovers the double-coset row in*

$$K\backslash G/H$$

*of every matching. Its linear span nevertheless has dimension only two: over $\mathbb{F}_{11}$ it is the plane*

$$2a + 2b + c = 0, \qquad 9a + 8b + d = 0. \tag{6.9}$$

*The two singleton profiles recover their individual matchings. Under the matching-decorated parent correspondence, they therefore recover the corresponding individual $H_3$ parent. A profile in a fibre of size four or six recovers only that $K$-orbit, not an individual matching.*

*Proof mode.* The subgroup-mark reduction and the implications from equivariance are conceptual. The six representative incidence rows in (6.10) below are certificate-checked exact counts and are reconstructed by an independent implementation. A companion certificate verifies that the matching decoration uniquely recovers its parent on the 22-point $H_3$ locus.

*Proof.* The permutation character of $K$ on either eleven-point sheet has values

$$(11, 3, 2)$$

on elements of orders $1, 2, 3$. The marks of the transitive $A_4$-sets are

| $K$-set | size | marks on orders $1, 2, 3$ |
|---|---|---|
| $K/K$ | 1 | $(1, 1, 1)$ |
| $K/V_4$ | 3 | $(3, 3, 0)$ |
| $K/C_3$ | 4 | $(4, 0, 1)$ |
| $K/C_2$ | 6 | $(6, 2, 0)$ |
| $K/1$ | 12 | $(12, 0, 0)$. |

Comparing marks gives the unique nonnegative decomposition

$$X_+|_K \simeq K/K \ \sqcup \ K/C_3 \ \sqcup \ K/C_2.$$

The orbit sizes are therefore $1, 4, 6$; conjugation by $J$ gives the same decomposition on $X_-$.

For $k \in K$, the transformation $k$ preserves every cell in (6.6) and carries $S_M$ to $S_{kM}$. Thus $D(kM) = D(M)$. The involution $J$ exchanges the two cells in every oriented pair and carries $S_M$ to $S_{JM}$, so

$$D(JM) = -D(M).$$

It remains to count one representative on each positive-sheet orbit. The exact incidence rows are

$$\begin{array}{cccc} \text{orbit size} & \mathrm{Stab}_K(M) & \text{zero counts in the four pairs of (6.6)} & D(M) \\ \hline 1 & A_4 & (0, 6;\ 0, 0;\ 12, 0;\ 0, 12) & v_1, \\ 4 & C_3 & (3, 6;\ 3, 0;\ 3, 3;\ 6, 3) & v_2, \\ 6 & C_2 & (3, 0;\ 1, 3;\ 2, 4;\ 6, 6) & v_3. \end{array} \tag{6.10}$$

The $J$-images supply the three negative rows. These six vectors are pairwise distinct, so their fibres are precisely the six $K$-orbits.

Row reduction over $\mathbb{F}_{11}$ gives rank two and the equations (6.9). This rank drop does not merge orbit labels: linear rank and set-theoretic separation are different assertions. Finally, a singleton fibre contains its matching without residual choice. The stabilizer of the matching $M_A$ from (6.3) is the same $A_5$ as the stabilizer of its parent $A$. The equivariant decorated transform between the two $G/H$-orbits is therefore a bijection, so the matching also determines its parent. The stated limitation for the other four rows follows from their fibre sizes. □

**Corollary 6.2** (Decorated sheet classifier)**.** *Relative to the orientation in (6.6), the profile $D(M)$ determines the sheet containing $M$. The antipodal singleton pair $\{v_1, -v_1\}$ recovers the unordered golden matching pair and hence the unordered golden parent pair; choosing the orientation selects one member.*

*Proof.* The positive and negative profile sets in (6.8) are disjoint, and the only singleton fibres are $v_1$ and $-v_1$. Apply the matching–parent bijection from Theorem 6.1. □

This pointwise classifier does not contradict the failure of a linear sheet sign in Section 5. The map $D$ uses zero incidences with the selected scalar-$K$ refinement; it is not a linear functional on the affine quotient configuration. Without that extra decoration, the quadratic and cubic moment statements remain the intrinsic recovery mechanism.

The profile compression also retains the cubic-first orientation from Section 5. The three positive vectors satisfy the integral weighted relation

$$v_1 + 4v_2 + 6v_3 = 0. \tag{6.11}$$

**Corollary 6.3** (Orbit weights from normalized profiles)**.** *The three incidence-normalized vectors $v_1, v_2, v_3$ recover the orbit-size multiset $\{1, 4, 6\}$. They also recover the stabilizer orders $\{12, 3, 2\}$ inside $K \simeq A_4$. The underlying rational rays alone do not recover these weights.*

*Proof.* The coordinates of each $v_i$ are fixed by the cardinality differences in (6.7), so independent rescaling is not allowed. These three integral vectors span a plane over $\mathbb{Q}$, and (6.11) is the primitive positive generator of their relation lattice. Its coefficients recover the three orbit sizes without labels. Orbit–stabilizer then gives $12/1, 12/4, 12/6$. If the vectors are replaced by primitive generators of their rays, the relation becomes $1 : 2 : 1$, which proves the final warning. □

**Lemma 6.4** (Three-ray cubic forcing)**.** *Let $k$ have characteristic different from 2 and 3. Suppose $u_1, u_2$ form a basis of a two-dimensional $k$-space and $n_1, n_2, n_3 \in k^\times$ satisfy*

$$n_1u_1 + n_2u_2 + n_3u_3 = 0.$$

*For the signed antipodal measure*

$$\xi = \sum_{i=1}^{3} n_i(\delta_{u_i} - \delta_{-u_i}),$$

*the moments of degrees one and two vanish, while the cubic moment is nonzero.*

*Proof.* The relation kills the first moment, and antipodality kills every even moment. Write

$$u_3 = -\frac{n_1}{n_3}u_1 - \frac{n_2}{n_3}u_2.$$

In $\sum_i n_i u_i^{\odot 3}$, the coefficient of $u_1^{\odot 2} \odot u_2$ is

$$-\frac{3n_1^2 n_2}{n_3^2},$$

which is nonzero. The signed cubic is twice this tensor and is therefore nonzero as well. $\square$

**Corollary 6.5** (The mass-zero cubic flag)**.** *In Lemma 6.4, suppose also that $n_1+n_2+n_3 = 0$. Then its half-cubic factors as*

$$\frac{1}{2}M_3(\xi) = \sum_{i=1}^{3} n_i u_i^{\odot 3} = \frac{n_1 n_2}{(n_1+n_2)^2}(u_1 - u_2)^{\odot 2} \odot ((2n_1+n_2)u_1 + (n_1+2n_2)u_2). \quad (6.12)$$

*The two displayed lines are distinct. Hence the cubic intrinsically defines a doubled line and a residual simple line; its Hessian recovers the doubled line.*

*Proof.* Since $n_3 = -(n_1+n_2)$, the weighted vector relation gives

$$u_3 = \frac{n_1 u_1 + n_2 u_2}{n_1 + n_2}.$$

Substitution and expansion give (6.12). The residual factor would equal the doubled factor only if $3(n_1+n_2) = 0$, which is impossible under the hypotheses. The Hessian of a binary cubic $L^2R$, with $L$ and $R$ distinct, is a nonzero scalar multiple of $L^2$. $\square$

If

$$\eta = \delta_{v_1} + 4\delta_{v_2} + 6\delta_{v_3} - \delta_{-v_1} - 4\delta_{-v_2} - 6\delta_{-v_3},$$

then its signed symmetric moments obey

$$M_j(\eta) = (1-(-1)^j)\sum_{i=1}^{3} n_i v_i^{\odot j}, \qquad (n_1,n_2,n_3) = (1,4,6). \quad (6.13)$$

Equation (6.11) kills the first moment, and antipodality kills every even moment. The first two profiles are independent over $\mathbb{F}_{11}$, since the determinant of their first two coordinates is $-18 \neq 0$. Lemma 6.4 therefore forces the cubic to be nonzero without a coordinate calculation. In the frozen normalization, its first coordinate is

$$2((-6)^3 + 4(-3)^3 + 6(3)^3) = 6 \neq 0 \quad \text{in } \mathbb{F}_{11}. \quad (6.14)$$

Thus the six-row quotient preserves the first nonzero signed moment even though its linear image is only a plane. It is a set-separating realization of the mixed $K$–$H$ double-coset space, not a faithful linear quotient of its six-dimensional function space.

*Remark* 6.6 (The $H_3$ cubic flag). Here $1+4+6 = 0$ in $\mathbb{F}_{11}$, so Corollary 6.5 forces the double-line factorization. In the basis $e_1 = v_2, e_2 = v_3$ of the profile plane, the projective cubic is represented by

$$f(x,y) = x^3 + 7x^2y + 5xy^2 + 9y^3 = (x-y)^2(x-2y).$$

Its Hessian is

$$7x^2 + 8xy + 7y^2 = 7(x-y)^2.$$

Thus the cubic divisor intrinsically distinguishes a doubled line and a residual simple line, and the Hessian recovers the doubled line. This is an internal flag of the decorated profile plane. It does not supply a canonical map from the full relative-cubic space to that plane, nor does the six-profile quotient carry a descended $G^+$-action.

# 7 Modular depth and arithmetic gluing

Two changes of characteristic meet in the $H_3$ matching table. The first is modular: because a sheet has eleven points, its constant vector lies in the augmentation submodule, and the depth map kills precisely that vector. The second is arithmetic: the quadratic marker that produces the two $H_3$ frames splits over $\mathbb{F}_{11}$, while its $A_3$ analogue is inert over $\mathbb{F}_5$. The two statements use the same sheets, but they are not two descriptions of one quotient.

## 7.1 The depth plane as a modular quotient

Retain the notation of Section 6, put

$$k = \mathbb{F}_{11}, \qquad L = G^+ = \mathrm{PSL}_2(11), \qquad V = k[X_+] \simeq \mathrm{Ind}_H^L \mathbf{1},$$

and regard $V$ as the permutation module on one eleven-point sheet. Let $e_1, e_4, e_6$ be the indicator functions of the three $K$-orbits, in orbit-size order. Thus

$$V^K = ke_1 \oplus ke_4 \oplus ke_6, \qquad \mathbf{1} = e_1 + e_4 + e_6. \tag{7.1}$$

Linearizing the incidence construction in (6.7) gives

$$\mathcal{D} : V^K \longrightarrow k^4, \qquad e_i \longmapsto i\, u_i, \tag{7.2}$$

where $u_1 = v_1$, $u_4 = v_2$, and $u_6 = v_3$ are the profiles on the orbits of the indicated sizes. Equation (6.11) says exactly that $\mathcal{D}(\mathbf{1}) = 0$.

**Proposition 7.1** (Modular depth quotient)**.** *The $L$-module $V$ is the projective cover $P(\mathbf{1})$ and has Loewy dimensions*

$$1 \mid 9 \mid 1.$$

*The map $\mathcal{D}$ has kernel $k\mathbf{1}$ and induces an isomorphism*

$$P(\mathbf{1})^K / \operatorname{soc} P(\mathbf{1}) \xrightarrow{\;\sim\;} \langle u_1, u_4, u_6 \rangle_k. \tag{7.3}$$

*The sheet exchanged by $J$ carries the negative copy of this quotient.*

*Proof mode.* Projectivity, indecomposability, fixed-point exactness, and the socle quotient are conceptual. Absolute irreducibility of the nine-dimensional middle layer and the rank and kernel of $\mathcal{D}$ are exact generator-matrix calculations in the `relative-cubic-depth` bundle, reconstructed by its independent replay.

*Proof.* The order of $H \simeq A_5$ is prime to 11, so its trivial module is projective and induction makes $V$ projective. The action of $L$ on $X_+$ is doubly transitive. Hence $\mathrm{End}_L(V)$ is generated by the identity and the all-ones operator $E$. In characteristic eleven,

$$E^2 = 11E = 0,$$

so this endomorphism algebra is local. Thus $V$ is indecomposable. Its trivial head occurs once, and consequently $V = P(\mathbf{1})$. The constant vector is its one-dimensional socle. The remaining nine-dimensional layer is absolutely irreducible; equivalently, the exact generator matrices on it span the full 9-by-9 matrix algebra. This gives the displayed Loewy dimensions.

Taking $K$-fixed points is exact because $11 \nmid |K| = 12$. The orbit sizes $1, 4, 6$ give (7.1). The three vectors in (7.2) span the plane (6.9), and their only relation is (6.11), so $\ker \mathcal{D} = k\mathbf{1}$. This proves (7.3). Finally $D(JM) = -D(M)$, which attaches the outer character on the opposite sheet. $\square$

This quotient identifies the precise modular loss in the six-profile construction. Before linearization, the six profiles remain distinct and recover all six double-coset labels. After linearization, the constant orbit trade dies and only the two-dimensional depth plane remains. Conversely, the three rational rays in that plane remain distinct but do not retain their orbit weights. The weights $1, 4, 6$ are recovered only when the quotient is supplied with the incidence-normalized vectors of (6.8), as in Corollary 6.3.

*Remark* 7.2 (Reduction of the profile table). The four-coordinate profiles in (6.8) are integral before reduction. The greatest common divisor of their nonzero 2-by-2 minors is 3. Their span therefore drops below rank two only in characteristic 3. The six signed rows remain distinct away from characteristics 2 and 3: characteristic 2 identifies antipodes, while characteristic 3 sends the rows $v_1$ and $v_2$ to zero. This is a statement about the displayed integral table, not a construction of the marked conic geometry in those characteristics.

## 7.2 Split markers and the $H_3$ hinge

The arithmetic representatives of the three configurations come in conjugate pairs. Besides the base matchings in (4.1), we use

$$\begin{aligned} M_{B,-} &= \{\{0,\infty\},\{1,3\},\{2,6\},\{4,5\}\}, \\ M_{B,+} &= \{\{0,\infty\},\{1,5\},\{2,3\},\{4,6\}\}, \\ M_{H,-} &= \{\{0,10\},\{1,\infty\},\{2,7\},\{3,5\},\{4,8\},\{6,9\}\}. \end{aligned} \tag{7.4}$$

The pair for $A_3$ consists of two copies of $M_{A_3}$, the pair for $B_3$ is $(M_{B,-}, M_{B,+})$, and the pair for $H_3$ is $(M_{H_3}, M_{H,-})$. To make the arithmetic bridge part of the definition, let $\bar{k}$ be an algebraic closure of the working field and define the root-specialization maps

$$\begin{array}{ccc} T & \text{root} & \sigma_T(\text{root}) \\ \hline A_3 & \sqrt{2},\ -\sqrt{2} & M_{A_3},\ M_{A_3} \\ B_3 & 3,\ 4 & M_{B,-},\ M_{B,+} \\ H_3 & 4,\ 8 & M_{H_3},\ M_{H,-}. \end{array} \tag{7.5}$$

Thus the *marker datum* is the quadratic algebra together with $\sigma_T$, not the polynomial alone. Exact projective transport places the displayed $B_3$ and $H_3$ values in $\Omega_{B_3}$ and $\Omega_{H_3}$. The certificate also verifies that $\sigma_T$ is the reduction of the corresponding integral Coxeter frames; the theorem below needs only the explicit map (7.5).

**Lemma 7.3** (Split–inert frame calculation)**.** *The three marker polynomials factor as follows:*

$$\begin{array}{ccc} A_3,\ q=5 & t^2-2 & \textit{irreducible}, \\ B_3,\ q=7 & t^2-2=(t-3)(t-4) & \textit{split}, \\ H_3,\ q=11 & t^2-t-1=(t-4)(t-8) & \textit{split}. \end{array} \tag{7.6}$$

*In the first row the two geometric roots form one Frobenius orbit over $\mathbb{F}_{25}$. In each split row the nontrivial involution of the quadratic marker algebra exchanges the two rational roots.*

*Proof.* The nonzero squares modulo 5 are 1 and 4, so 2 is not a square. If $u^2 = 2$ in $\mathbb{F}_{25}$, the nontrivial $\mathbb{F}_5$-automorphism sends $u$ to $-u$, and hence exchanges the two roots. The remaining factorizations follow by direct substitution. □

**Theorem 7.4** (Rank-three arithmetic gluing)**.** *For these arithmetic representatives of the rank-three matching configurations, the following hold.*

(a) *The $A_3$ marker datum is fused over $\mathbb{F}_5$. The $B_3$ and $H_3$ marker data split into two rational matching sheets over $\mathbb{F}_7$ and $\mathbb{F}_{11}$, respectively.*

(b) *In the $H_3$ case, the stabilizers $H_+$ and $H_-$ of the singleton matching in each rational sheet are isomorphic to $A_5$, satisfy*

$$H_+ \cap H_- = K \simeq A_4, \qquad \langle H_+, H_- \rangle = \mathrm{PSL}_2(11), \tag{7.7}$$

*and are exchanged by a projectivity $J$ of nonsquare determinant. The normalizer $N_G(K)$ is a rational $S_4$ whose determinant-square kernel is $K$.*

(c) *The determinant character that distinguishes the two rational sheets also negates the depth profiles. Thus choosing one root in the split marker algebra chooses one sheet, while its singleton profile chooses the corresponding decorated matching row. Forgetting that choice leaves the unordered pair.*

*Proof mode.* Lemma 7.3 and the determinant-character deductions are conceptual. The agreement of the specialization map (7.5) with the integral Coxeter frames, the matching stabilizers and intersections, and the generated subgroup in (7.7) are exhaustive exact finite checks. The classical subgroup names use the standard subgroup structure of $\mathrm{PGL}_2(11)$ [1, 4].

*Proof.* By definition, the two roots in the $A_3$ row of (7.5) have the same matching value. The two values in the $B_3$ row are exchanged by $x \mapsto -x$, and the two values in the $H_3$ row are exchanged by

$$x \longmapsto \frac{x-1}{x+1}.$$

The exact transport check verifies these identities on the displayed matchings. Together with Lemma 7.3, they prove (a).

For $H_3$, exact projective closure gives

$$|H_+| = |H_-| = 60, \qquad |H_+ \cap H_-| = 12, \qquad |\langle H_+, H_- \rangle| = 660.$$

Both stabilizers lie in the determinant-square subgroup, while the displayed transporter has nonsquare determinant. The normalizer $N_G(K)$ has order 24, and its determinant-square kernel has order 12. The cited subgroup identifications give (7.7) and the $S_4/A_4$ hinge, proving (b).

The determinant-square subgroup has the two matching sheets as its orbits, and $J$ exchanges them. Section 6 proves $D(JM) = -D(M)$ and identifies the unique singleton in each sheet. Theorem 6.1 then returns the decorated matching row from that singleton. This proves (c). □

Arithmetic splitting and modular depth retain different amounts of information. Splitting produces a two-point choice of sheet. The depth quotient is defined only after the $A_4$-refinement has been selected and remembers all three orbit rays modulo the constant socle. The first choice orients the second, but does not canonically identify it with any cubic quotient of the ten-dimensional factorization space. Appendix A makes that boundary precise.

## 8 Verification architecture

Table 1 separates the mathematical arguments from their exact finite inputs. Here "certificate" means a deterministic calculation over a prime field, checked against a canonical JSON output and a SHA-256 manifest; an independent replay uses a separate implementation. "Lean" means that the stated finite implication is checked by the Lean kernel. Neither term replaces the cited identification of the classical groups or configurations.

| Result | Proof in the text or classical input | Exact evidence |
|---|---|---|
| Matching-secant quotient | unique conic divisibility and the four-endpoint switch | none |
| Ranks and the middle Fischer layer | harmonic decomposition; Edge–Dye configurations | quotient matrices and independent replay |
| Balanced sheets and cubic orientation | Proposition 5.1; tensor identities | exact moments; radical hypotheses and independent replay |
| Six-profile reconstruction | subgroup marks, orbit–stabilizer, and the three-ray lemmas | profile rows and replay; decorated-parent bijection |
| Modular depth and the Tate plane | projectivity, fixed-point exactness, and standard modular-module input | invariant, coinvariant, and depth calculations with replay |
| Arithmetic splitting and gluing | displayed root calculations; cited subgroup identifications | Lean gate and normalized certificate/replay |

Table 1: Proof modes for the principal results.

The claim-by-claim map is `verification/trust_manifest.json`; the exact sixteen-statement identity is `verification/statement_identity.json`. From the root of the archival source, the aggregate entry point is

```
python3 papers/clebsch-factorization/verification/verify_release.py
```

It checks the statement identity and all six checksum manifests, runs the admitted primary checks and independent replays, and rebuilds this manuscript through the repository Makefile.

The full-source review archive accompanying this candidate is pinned by `verification/evidence_fingerprint.json`, whose SHA-256 digest is

`2d16e063d7ed7480adebec277179dc39`
`c6aa7642a1a6bdb116bc3c7324550a2e`.

That fingerprint fixes the six evidence manifests, command vectors, runner, trust manifest, and environment: Python 3.13.12, Lean 4.32.0-rc1, and Mathlib revision

`571b8a8e54219b4d393f75f4b8653fac08197fcc`.

A successful replay ends with

`clebsch factorization release: CHECK OK`.

An immutable public archive locator must be added before release. Until then, the digest identifies the accompanying review archive exactly but is not presented as a public deposit.

The largest formal component is the gate

`RelativeConicArcs.Gates.`
`ClebschArithmeticGluing`.

Its twenty-three terminals check the displayed roots, matching involutions, projective orders, determinant halves, the $A_3/B_3$ orbit calculations, and the split–fused conclusion. For $H_3$, Lean checks the normalized certificate rows, intersections, determinant classes, the $11 + 11$ signature split, and the shape of the bounded generation words. Evaluation and completeness of the 660-element coset and word certificates remain exact generator/replay obligations. The classical names $S_4, A_4, A_5$, and $\mathrm{PSL}_2(11)$ are cited inputs: no group name is inferred from an order.

The other computations are load-bearing only at the points shown in Table 1. The matching-module bundle supplies the three finite quotient ranks and frozen moment data;

the balanced-sheet bundle checks the finite hypotheses of the symbolic radical argument; the profile-incidence bundle computes one row per double coset; the decorated-parent bundle identifies a singleton matching with its parent; and the relative-cubic-depth bundle computes the modular and Tate rank patterns. All use exact prime-field arithmetic. The general divisibility theorem, radical–Hadamard lemma, three-ray cubic lemma, and mass-zero factorization are proved in the text and do not depend on those enumerations.

This verification surface proves only the three marked configurations defined here. It does not supply an all-characteristic construction, a canonical map from the relative-cubic Tate plane to the depth plane, or any passage, holonomy, theta, Fourier, quantum, or Mathieu identification.

# A The relative-cubic Tate plane

Let $W = W_{H_3}$ be the ten-dimensional quotient space from Theorem 4.1, let $\chi$ be the determinant character of $G = \mathrm{PGL}_2(11)$, and put

$$M = \mathrm{Sym}^3(W) \otimes \chi, \qquad R = M^G.$$

The signed cubic of Section 5 lies in $R$, which has dimension three. There is a canonical two-dimensional quotient of this source space, but it is not canonically the depth plane of Proposition 7.1.

**Proposition A.1** (Canonical relative-cubic Tate plane)**.** *The invariant space $R$ and the coinvariant space $M_G$ both have dimension three. The canonical projection and group norm fit into*

$$R \xrightarrow{\pi} M_G \xrightarrow{N} R, \qquad \mathrm{rank}\,\pi = 2, \quad \mathrm{rank}\,N = 1, \tag{A.1}$$

*with*

$$\mathrm{im}\,\pi = \ker N, \qquad \mathrm{im}\,N = \ker \pi = \langle [1:3:9] \rangle. \tag{A.2}$$

*Contraction by the covector whose square is the unique rank-one member of the invariant symmetric-form pencil gives a second construction of the same quotient. In the frozen relative basis its matrix is*

$$\begin{pmatrix} 8 & 1 & 0 \\ 0 & 8 & 1 \end{pmatrix}, \tag{A.3}$$

*and its kernel is again $\langle [1:3:9] \rangle$.*

*Proof mode.* The invariant/coinvariant construction and the implications of the displayed ranks are conceptual. The dimensions, norm and projection ranks, kernel line, rank-one pencil member, contraction matrix, and flattening census are exact finite linear algebra in the `relative-cubic-depth` bundle and its independent replay.

*Proof.* Exact invariant and commutator calculations give $\dim R = \dim M_G = 3$, the two ranks in (A.1), and $N\pi = 0$. Thus $\mathrm{im}\,\pi \subseteq \ker N$; both spaces have dimension two, so they are equal. The computed image of $N$ is the displayed kernel of $\pi$, proving (A.2).

The invariant symmetric-form pencil on $W$ has a unique rank-one member $\ell^2$, where $\ell$ transforms by $\chi$. Contracting a $\chi$-twisted cubic by $\ell$ cancels the two characters and lands in the invariant quadratic pencil. Exact contraction gives (A.3), whose kernel proves the second assertion. □

The kernel line is intrinsic rather than an artifact of the displayed basis. Among the 133 projective lines of $R$, it is the unique line whose first flattening has rank 9; the remaining census is one line of rank 1 and 131 lines of rank 10.

*Remark* A.2 (Why the two canonical planes do not glue). The three labelled cubic orbit classes project to $M_G$ with relation $[2, 9, 1]$, whereas the three labelled depth profiles have relation $[2, 8, 1]$. There is therefore no label-preserving linear map carrying these source classes to the depth rays. The integral transfer makes the obstruction structural. In orbit-size order set

$$s = (1, 1, 1)^t, \qquad B = s(1, 4, 6).$$

Then $B^2 = 11B$. Divided transfer kills every integral balanced relation $a$ with $(1, 4, 6)a = 0$, but $Bs = 11s$, so it fixes the depth socle after division by 11. It separates the source relation from the depth socle instead of identifying them. The doubled-line and residual-line flag of Remark 6.6 is internal to the depth plane and supplies no missing map.

## Conclusion

Restriction to a conic erases the pairing of marked points, but division by its equation does not erase every trace of the factorization. The four-endpoint switch turns the competing secant products into an affine quotient configuration. For the rank-three systems this configuration has dimensions $3, 6, 10$: its top harmonic part always survives, and in the even case so does the deepest radial line. In type $H_3$, the omitted information is exactly the middle Fischer layer $Q\mathcal{H}_2$.

The quotient remembers more than its linear span. In the balanced $B_3$ and $H_3$ cases, quadratic products recover the unordered pair of sheets, while the first signed tensor that survives is cubic and changes sign under the outer involution. After an $A_4$ refinement has been chosen, six incidence profiles still separate all six matching rows, although their linear image is only a plane. Over $\mathbb{F}_{11}$ the same incidence-normalized weights $1, 4, 6$ describe both the rational orbit sizes and the loss of the constant socle in modular depth; the three projective rays alone do not determine those weights.

These recovery statements have precise limits. Balanced moments do not order the sheets or choose the $A_4$ refinement. The nonsingleton profiles recover orbits, not individual matchings. The relative-cubic Tate plane and the depth plane are each canonical relative to their stated inputs, but the natural labelled and transfer maps do not identify them. Arithmetic splitting chooses a rational sheet in the three frozen small-field models; it does not produce an all-prime reciprocity law.

Thus the conic ideal retains a layered memory: harmonic and radial linear data, an unordered balanced-sheet decomposition, cubic orientation, and—after a stated refinement—the exact matching-table label. Pointed passage maps and global holonomy require additional structure, so they lie beyond the reconstruction theorem proved here.

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